Do More Text Views Help? A Leakage-Aware Evaluation of Text Representations for Sequential Recommendation

Anonymous Author(s)

Abstract

Text features are widely used to enrich sequential recommenders, but comparisons of lexical TF-IDF against TF-anchored LLM augmentations (single-view TF+LLM or multi-prompt TF+MV) are often conducted under evaluation protocols that permit temporal overlap. We conduct a leakage-aware controlled study of ID-only, cutoff-fit TF-IDF, TF+LLM (a single Qwen2.5-7B view added to the TF-IDF base), and TF+MV (four prompt views added to the TF-IDF base) under full-ranking global temporal splitting (GTS) on three Amazon datasets, using GTS-Last as the primary protocol, GTS-Random as a robustness check, and clean leave-one-out (LOO) as a contrast. Three findings emerge. First, evaluation protocol changes conclusions: across the evaluated settings, text consistently improves over ID-only on warm targets, but the relative ordering of TF-IDF, TF+LLM, and TF+MV changes across Last, Random, and LOO, even when Last and Random evaluate the same trained checkpoints. Second, no text representation is robustly dominant across temporal windows; temporal shift reduces absolute performance while preserving near-ties among text variants. Third, additional prompt views provide limited marginal value because they are highly redundant: inter-view CKA reaches 0.82–0.94, the concatenated representation uses only 53–55% of its nominal effective rank, and leave-one-view-out has negligible downstream effect. In contrast, TF-IDF is geometrically distinct from LLM views (CKA $\approx$ 0.19), suggesting a potentially complementary lexical–semantic axis that is not captured by adding more prompt views. These results suggest that evaluation protocol and representation redundancy should be diagnosed before attributing gains to increasingly complex LLM text pipelines.

CCS Concepts

• **Information systems** → **Recommender systems**; *Content ranking*.

Keywords

Sequential Recommendation, Text Features, Temporal Evaluation, Multi-View Redundancy, Leakage-Aware Protocol

1 Introduction

Sequential recommendation predicts a user's next item from historical interactions [6, 10, 21]. Item text—titles and descriptions—is a standard way to enrich sequential recommenders and enable cross-domain transfer [8, 13]. A natural working hypothesis is that *richer text always helps*: pretrained language embeddings should beat TF-IDF, and multiple prompt-based views of the same item should beat a single view. Recent work shows that pretrained language representations can improve text-enhanced sequential recommendation and transferability [8, 13], and that feature-level attention can exploit item attributes [25]. We ask whether multiple prompt-conditioned views of the same item text—in this study, description, function, audience, and style—add complementary signal beyond a single view. That comparison, and whether cutoff-fit TF-IDF already suffices, depends on how evaluation is set up.

This hypothesis rests on fragile evaluation foundations. Most text-enhanced sequential recommenders are validated with leave-one-out (LOO) splits that, as shown by Ji et al. [9] and Sun [20], allow temporal overlap between training and test, yielding higher absolute metrics and potentially distorting model comparisons. In a deployment scenario, a recommender must predict items that a user will interact with *in the future*, not reconstruct a randomly held-out historical interaction. Global temporal split (GTS) better simulates this deployment constraint, but introduces additional design choices: the *target-selection mode* (Last, Random, Successive, or All interactions per user in the test window) itself changes effective task difficulty and can alter model rankings [4]. Prior work has studied split effects in general [9, 15]; we show that these effects are *specifically decisive* for the TF-IDF vs. TF+LLM vs. TF+MV comparison.

Text features also require preprocessing—vocabulary construction, IDF weighting, SVD projection, and LLM embedding whitening—each of which must respect the temporal boundary. If these statistics are computed on the full item set (including future items), the resulting features carry information unavailable at training time, introducing a form of leakage orthogonal to the train/test split itself. We therefore enforce strict *cutoff-fit* controls throughout: all text preprocessing is fit only on items appearing before the training cutoff.

We compare five variants—ID, TF-IDF, TF+LLM, TF+MV, and an adapted UniSRec baseline [8]—on three Amazon datasets under GTS-Last (primary), GTS-Random (robustness check), and clean LOO (contrast), plus an equal-horizon Early temporal window. We name variants by their full feature composition: the LLM-based variants *augment* the lexical TF-IDF base rather than replace it; a base-free LLM-only variant is evaluated separately in Section 7. We diagnose multi-view behavior with CKA [11], effective rank [19], and leave-one-view-out (LOVO).

Research questions.

RQ1 *Protocol effect.* Do clean LOO, GTS-Last, and GTS-Random change the absolute gains and relative ranking of TF-IDF, TF+LLM, and TF+MV?

RQ2 *Temporal robustness.* Can the conclusions drawn under the Main temporal window (80% cutoff) survive when the training window shrinks (Early, 60% cutoff)?

RQ3 *Multi-view redundancy.* Why do four prompt views (TF+MV) not reliably outperform a single view (TF+LLM) or even TF-IDF alone, and where does potential complementarity lie?

Contributions.

(1) A leakage-aware comparison on three datasets showing that evaluation protocol changes text-representation rankings. Within GTS, Last vs. Random on the same trained checkpoints yields different winners (**target effect**); switching from GTS to LOO changes both training data and targets (**split effect**). Text consistently beats ID, but *which* text is best is protocol-sensitive.
(2) Cross-window evidence that no single text variant dominates: under a matched-protocol equal-horizon Early window on Beauty (60% cutoff; the same 10% valid/test spans as Main, final 20% unused), absolute MRR drops and the Main-window ranking is not preserved, yet text variants remain within 0.001 of each other.
(3) A triangulated redundancy analysis (CKA + effective rank + LOVO) spanning raw 3584-d Qwen hidden states through 64-d post-whitening, jointly explaining limited multi-view gains, while identifying lexical–semantic dissimilarity (TF vs. LLM, CKA 0.18–0.20) as a potential complementarity axis.

2 Related Work

Text-enhanced sequential recommendation. Early systems fused item-content bag-of-words with collaborative filtering [23]. Modern approaches encode titles and descriptions with pretrained language models for cross-domain transfer [7, 8], or use language models as the recommender itself [1, 3]. UniSRec learns transferable item representations from text and adapts them via a lightweight MoE [8]; VQ-Rec quantizes textual embeddings for transfer [7]; P5 casts recommendation as language modeling [3]; TALLRec aligns LLMs to ranking via instruction tuning [1]. These systems typically report LOO or loosely specified temporal protocols. We do *not* propose a new architecture; we ask whether adding LLM views—and whether adding more of them—remains advantageous once temporal leakage is controlled under a fixed backbone.

LLM-based item representations. Beyond architecture papers, language-model embeddings can also be used as drop-in item features, either from a single title/description encoding or from several prompt-conditioned views of the same text. This paper studies the second choice: whether prompts aimed at different facets (function, audience, style) yield complementary signal. The motivating intuition is that different prompts can elicit diverse semantic aspects of an item (e.g., "describe the function" vs. "describe the target audience"), thereby increasing feature coverage. However, when all views are produced by the *same* pretrained encoder applied to the *same* item text, the diversity of the resulting representations depends on how much the encoder's internal representations actually change in response to different instructions—a question that has received little empirical attention in the recommendation literature. Our study isolates this representation choice from architectural novelty: the same SASRec backbone and fusion pipeline are held fixed while only the text source changes (TF-IDF alone, TF+LLM, or TF+MV).

Evaluation protocols and temporal leakage. Sampled metrics distort rankings [12]; LOO vs. time-based splits change absolute scores and can alter model selection [9, 15, 20]. Hidasi and Karatzoglou [5] showed that seemingly minor protocol choices can cause large shifts in relative rankings of sequential models. Within GTS, target-selection mode matters: Gusak et al. [4] show that Successive best approximates online next-item prediction but is expensive; Last is a deterministic low-cost proxy; Random has the highest rank-correlation with Successive (we average three target-sampling seeds because Random is non-deterministic); and All packs multiple items into one ground-truth set, introducing task mismatch and the weakest correlation with Successive. Beyond sequential recommendation, the impact of temporal protocols has also been documented in session-based [5] and general collaborative filtering settings [15], confirming that protocol sensitivity is not a niche issue but a systemic concern across recommendation paradigms. We specialize this finding to text representations: protocol changes conclusions *precisely because* TF-IDF, TF+LLM, and TF+MV respond differently to distribution shift. Matching validation and test target modes (both Last, or both Random) is also recommended for reliable model selection [4]; our Early-window experiments keep this alignment.

Multi-view learning and representation redundancy. Multi-view theory requires complementarity [24]. In computer vision, Kornblith et al. [11] introduced CKA (Centered Kernel Alignment) to measure representational similarity of Gram structures across network layers and architectures. In NLP, Ethayarajh [2] revealed that contextual word embeddings from different layers of the same pretrained model are highly anisotropic and occupy a narrow cone in embedding space. Effective rank (the exponential of the entropy of normalized singular values) [19] provides a complementary diagnostic, quantifying how much of a representation's nominal dimensionality is actually used. CKA has been used to compare layers and architectures [11], but to our knowledge CKA and effective rank have not been jointly deployed to diagnose *multi-prompt* redundancy in recommendation. We combine CKA, effective rank, and LOVO—not any single measure alone—to explain limited multi-view gains.

3 Controlled Study Setup

3.1 Temporal Protocols

GTS-Last (primary). Interactions are globally ordered by timestamp (RecBole TS [26]). Main uses the first 80% for training, the next 10% for validation, and the last 10% for testing. Early uses the first 60% for training, the next 10% for validation, and the next 10% for testing; the final 20% is unused, so the evaluation window is shifted earlier while validation/test horizon lengths remain fixed. After sequential augmentation, each user contributes only their chronologically last test interaction (one target per user)—the deterministic last-item proxy studied by Gusak et al. [4].

GTS-Random (robustness check). Same checkpoints; each user contributes one uniformly random test interaction. We average over three target-sampling seeds (42, 123, 456) for stability; cross-seed std per checkpoint is 0.001–0.003.

Clean LOO (contrast). Per-user leave-one-out with no global time cutoff: for each user, all but the final two interactions are used for training, the penultimate interaction for validation, and the final

interaction for test (RecBole LS split). Included only to quantify the protocol gap; *not* used for primary conclusions.

GTS-All (appendix). All post-cutoff targets per user; relegated to Appendix A due to task-mismatch concerns [4].

Leakage controls. TF-IDF vocabulary/IDF, SVD projection, and LLM center/whitening are fit only on items appearing before the training cutoff. Evaluation filters users with ≥ 5 training interactions. Main tables report *warm-target* results: the test target must belong to the training-time item catalog, which is also the candidate set at inference. Last and Random are evaluated on the *same eligible-user set* per dataset (users with at least 5 training interactions and a test-window interaction: Beauty 4,520, Toys 4,873, Grocery 4,152); the two modes differ only in which target interaction is selected. Because the warm filter applies to the *selected* target, the warm sub-cohort differs slightly between modes (e.g., Beauty: 3,137 warm users under Last vs. 3,054–3,083 across Random target-sampling seeds). Appendix D re-evaluates both modes on the intersecting warm users. Unseen-target shares and popularity buckets for the same cutoffs are in Tables 5–6.

3.2 Text Representations and Models

All variants share a SASRec [10] backbone (d=256, 2 layers, 2 attention heads).

- **ID**: ID embeddings only.
- **TF**: train-cutoff TF-IDF → SVD 256-d → lightweight projection head + InfoNCE alignment [22].
- **TF+LLM**: the TF-IDF base of TF *plus* a single Qwen2.5-7B [17] item embedding (train-cutoff centered and whitened); both sources are fused by the same pipeline.
- **TF+MV**: the TF-IDF base *plus* four prompt views (Description / Function / Audience / Style) → per-view 64-d projections → 256-d concatenation → shared projection.
- **UniSRec** [8]: the original UniSRec architecture (MoE adaptor over item-text embeddings) trained from scratch on each dataset with no cross-domain pretraining; for leakage parity, its item-text inputs are replaced by the same cutoff-fit whitened Qwen2.5-7B embeddings, under the identical evaluation filter. We therefore treat it as a *UniSRec backbone adapted to our cutoff-fit features*, not as the original pretrained UniSRec.

The variant names make the feature composition explicit: the LLM-based variants *augment* the lexical base rather than replace it, so RQ1 compares incremental augmentation of a fixed TF-IDF base, not three independent representation families. A base-free LLM-only variant is evaluated in Section 7. All text variants use two-phase training: Phase A (text-head warmup via InfoNCE), then Phase B (joint BPR [18] + InfoNCE fine-tuning). Hyperparameters and cutoff-fit preprocessing for the Early window are frozen from Main (no re-tuning).

3.3 Datasets

We evaluate on three Amazon review categories [14, 16]. Tables use the short names Beauty, Toys, and Grocery:

- **Beauty**: 1.21M users, 259K items, 2.02M interactions.
- **Toys & Games**: 1.28M users, 336K items, 2.25M interactions.
- **Grocery & Gourmet Food**: 768K users, 172K items, 1.30M interactions.

Table 1: Protocol contrast — warm-target MRR@10 (3 training-seed mean±std; users with ≥ 5 training interactions; GTS-Random additionally averaged over 3 target-sampling seeds). Bold: highest value in the row (exact ties both bold). UniSRec is an adapted external baseline and is strongest on Grocery. Among the SASRec-based variants: Last favors TF+MV on Beauty/Grocery (Toys TF+MV ≈ TF); Random: TF ≈ TF+MV; LOO: TF+LLM ≈ TF+MV ≈ TF.

Dataset	Protocol	ID	TF	TF+LLM	TF+MV	UniSRec
Beauty	GTS-Last	.0032±.0006	.0053±.0004	.0060±.0004	**.0064**±.0003	.0052±.0002
	GTS-Rand	.0074±.0005	**.01239**±.0002	.01201±.0002	**.01244**±.0002	.0096±.0001
	LOO	.0175±.0004	.0251±.0002	**.0254**±.0004	.0253±.0004	.0239±.0002
Toys	GTS-Last	.0031±.0002	.0040±.0005	.0035±.0002	**.0042**±.0006	.0040±.0008
	GTS-Rand	.0076±.0002	**.0113**±.0004	.0108±.0007	.0108±.0009	.0085±.0003
	LOO	.0152±.0005	.0247±.0006	**.0258**±.0012	**.0258**±.0028	.0196±.0010
Grocery	GTS-Last	.0030±.0012	.0046±.0006	.0047±.0005	.0058±.0007	**.0107**±.0002
	GTS-Rand	.0050±.0012	.0087±.0011	.0076±.0004	.0089±.0008	**.0121**±.0002

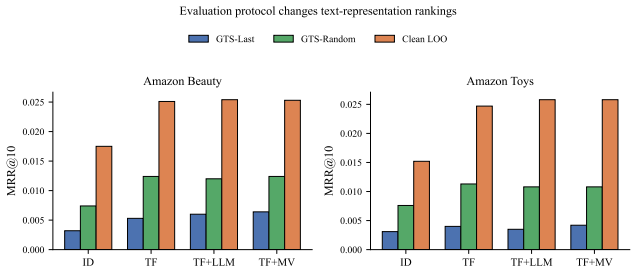


Figure 1: Protocol contrast on Beauty and Toys. GTS-Last/Random: warm MRR@10 (3 training seeds; Random further averaged over 3 target-sampling seeds); LOO is clean leave-one-out, shown for contrast only. Bar heights are *point estimates*; gaps within one seed-level std use ≈. Per-row orderings are in Table 1.

3.4 Evaluation Details

Full-ranking Hit@10, NDCG@10, MRR@10 [12]. Main tables: MRR@10, mean±std over 3 training seeds (42, 2024, 2026). GTS-Random further averages 3 target-sampling seeds (42, 123, 456) on those same checkpoints; NDCG and Hit are reported as secondary metrics. All runs on a single NVIDIA RTX 5090; per-model training: 15–65 min.

4 RQ1: Protocol Changes Rankings

Table 1 and Figure 1 report the protocol contrast. No text representation wins everywhere; instead, the relative ordering shifts with the target-selection protocol:

- **GTS-Last**: among the controlled SASRec-based variants, point estimates favor TF+MV on Beauty and Grocery (.0064 / .0058); on Toys TF+MV ≈ TF (.0042 / .0040), with seed-dependent winners

Table 2: GTS-Last warm NDCG@10 and Hit@10 (3-seed mean±std). Bold: highest value in the row (same rule as Table 1). Among the SASRec-based variants, NDCG agrees with MRR. UniSRec is strongest on Hit@10 (all three datasets) and on Grocery NDCG.

Dataset	Metric	ID	TF	TF+LLM	TF+MV	UniSRec
Beauty	NDCG	.0047±.0007	.0074±.0006	.0080±.0009	**.0087**±.0003	.0081±.0003
	Hit	.0098±.0013	.0146±.0013	.0146±.0029	.0167±.0011	**.0180**±.0007
Toys	NDCG	.0046±.0004	.0056±.0006	.0046±.0002	.0057±.0011	**.0066**±.0013
	Hit	.0098±.0013	.0108±.0011	.0084±.0004	.0108±.0027	**.0154**±.0030
Grocery	NDCG	.0048±.0019	.0062±.0004	.0063±.0005	.0077±.0007	**.0160**±.0003
	Hit	.0107±.0041	.0117±.0009	.0117±.0007	.0136±.0007	**.0338**±.0007

Table 3: Per-seed GTS-Last warm MRR@10 (seeds 42 / 2024 / 2026). Variance is visible; TF+MV's lead is small and seed-dependent.

Dataset	Seed	ID	TF	TF+LLM	TF+MV	UniSRec
Beauty	42	.0038	.0049	.00627	**.00629**	.0050
	2024	.0032	.0052	.0055	**.0066**	.0054
	2026	.0026	.0057	**.0063**	.0061	.0051
Toys	42	.0033	**.0045**	.0037	.0035	.0044
	2024	.0030	.0038	.0032	.0042	**.0045**
	2026	.0029	.0036	.0036	**.0048**	.0031

(Table 3); the adapted UniSRec baseline is strongest on Grocery (.0107).

- **GTS-Random** (3 target-sampling seeds): TF leads on Toys (.0113 vs. TF+MV .0108, within one std of TF+MV); Beauty is a tie (TF .01239, TF+MV .01244, both round to .0124); Grocery is a tie (TF .0087, TF+MV .0089).
- **LOO**: absolute MRR is 4–7× higher than GTS-Last; ranking becomes TF+LLM ≈ TF+MV ≈ TF.

Last vs. Random is a **target-selection** effect on the same trained GTS checkpoints; LOO vs. GTS is a **split/training-protocol** effect (different training data and models).

A plausible account of the LOO gap. The 4–7× higher absolute scores under LOO also change the relative ordering among text methods. LOO permits temporal overlap: a user's training interactions include items from the same time period as the test target, effectively allowing the model to "memorize" co-occurrence patterns that would not exist under a strict temporal cutoff. This overlap may disproportionately benefit representations capturing fine-grained semantic similarity (TF+LLM and TF+MV), since overlapping items provide richer co-occurrence signal for semantic features than for lexical ones. Under GTS, where training and evaluation interactions are separated by a global temporal cutoff, that LOO ranking is no longer visible and TF-IDF becomes competitive.

Why do Last and Random disagree? Under GTS-Last, each user's single evaluation target is their *chronologically last* test interaction; under GTS-Random, it is uniformly sampled from the test window. A text representation that "wins" under one target mode can "lose" under another, even with identical model weights. Last may over-represent recently popular or niche items while Random draws a broader mix; regardless of the cause, our claim is only that changing target selection on the same checkpoints changes the winner. This is not an artifact of the two modes scoring slightly different warm cohorts: restricting both to their intersecting warm users preserves the pattern (Appendix D).

Table 4: Main (80% cutoff) vs. Early (60% cutoff) on Beauty — GTS-Last warm MRR@10 (3-seed mean±std). Early: first 60% train, next 10% valid, next 10% test (final 20% unused). Fusion and cutoff-fit embeddings matched to Main; hyperparameters frozen. The Main row matches Table 1. Point estimates: TF+MV on Main, TF-IDF on Early (within one std of TF+MV).

Window	TF	TF+LLM	TF+MV
Main	.0053±.0004	.0060±.0004	**.0064**±.0003
Early60-EQ	**.0042**±.0005	.0037±.0003	.0039±.0003

Table 5: All-target catalog-level context (valid→test, Main GTS), *not* a direct explanation of warm-target MRR. Unseen targets are absent from the train catalog and never enter the warm scores of Tables 1 and 4. Reported as associations with catalog shift, not causal claims.

Dataset	Unseen targets	Pop mean (V→T)	Pop median (V→T)
Beauty	31.9%	112.8 → 101.8	21 → 15
Toys	40.8%	144.7 → 119.1	18 → 11
Grocery	29.6%	87.3 → 76.9	16 → 12

Implications. All text methods outperform ID-only under every protocol we evaluate (+13–100% relative under GTS-Last); absolute scores are higher under Random.

UniSRec as external baseline. Under the same GTS-Last protocol, users with at least five training interactions, and warm-target filtering, UniSRec scores 0.0052 (Beauty) and 0.0040 (Toys), between ID and TF on Beauty and tied with TF on Toys, but it leads on Grocery (0.0107, nearly 2× the next best). Table 2 shows that NDCG@10 largely agrees with MRR rankings under GTS-Last, while Hit@10 elevates UniSRec across all three datasets, so metric choice interacts with protocol. We therefore treat MRR as the primary ranking metric and report NDCG/Hit for completeness. UniSRec's apparent ranking is protocol-sensitive: GTS-All warm is .0180 / .0139 on Beauty / Toys, versus GTS-Last warm .0052 / .0040 on the same checkpoints (Appendix E). Target mode and candidate scope can each shift where an external baseline appears to rank.

5 RQ2: Temporal Robustness

Table 4 reports GTS-Last warm MRR@10 on Beauty for both windows under the same cutoff-fit protocol. Absolute scores drop from Main to Early for every variant. The within-window ranking is *not* invariant: point estimates favor TF+MV on Main and TF-IDF on Early (.0042 vs. TF+MV .0039 and TF+LLM .0037, within one std of TF+MV). All three remain within 0.001 of each other. TF-IDF also degrades less than TF+LLM/TF+MV when the cutoff recedes, so the

Table 6: All-target popularity-bucket composition (valid→test, Main GTS; head ≥10, mid 3–9, low 1–2 train interactions).

Dataset	Split	Head	Mid	Low	Unseen
Beauty	Valid	63.1%	15.9%	9.7%	11.3%
	Test	56.6%	14.9%	10.7%	17.8%
Toys	Valid	60.1%	16.3%	10.6%	13.0%
	Test	52.3%	15.5%	11.0%	21.3%

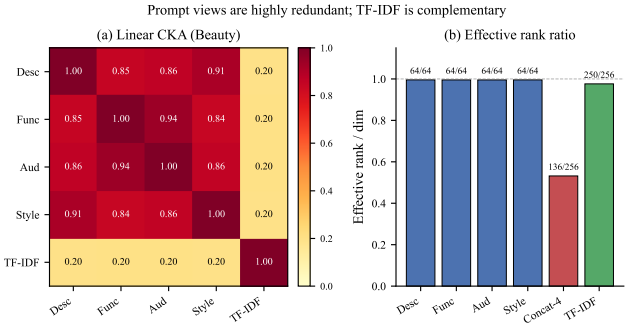


Figure 2: Representation analysis on Beauty (3 random item subsamples of 3,000 each, mean±std). (a) Pairwise CKA: LLM views show high mutual similarity (0.84–0.94); TF–LLM CKA is low (≈0.20). (b) Effective rank: 4-view concat 136/256 (53%) vs. TF-IDF 250/256 (98%).

Main-window "TF+MV slightly ahead" pattern does not survive as a temporal fact. The Main→Early drop in Table 4 is measured on warm targets only. Tables 5–6 provide all-target *catalog-level context* (unseen share, popularity) for the same global cutoffs; they are not a direct explanation of warm-MRR degradation, because unseen items never enter those scores.

TF-IDF robustness under shift. TF-IDF exhibits the smallest Main-to-Early drop among the three text variants (Table 4); we treat this as an empirical robustness observation rather than evidence for a causal mechanism.

6 RQ3: Why Limited Multi-View Gain

We address RQ3 with three complementary analyses, none of which is sufficient alone: CKA measures Gram-matrix similarity [11], effective rank measures dimensionality utilization, and LOVO measures downstream impact.

(1) High inter-view CKA.. Table 7 and Figure 2(a) show that LLM view pairs have CKA 0.82–0.94 (Beauty 0.84–0.94; Toys 0.82–0.92, except Function–Style .818). The four prompt views, generated from the same encoder applied to the same item text with different instructions, encode highly similar relational structures. In contrast, TF-IDF vs. any LLM view yields CKA 0.196–0.201 (Beauty) and 0.184–0.189 (Toys)—a qualitatively different regime indicating substantial representational dissimilarity.

Table 7: Inter-view CKA (3 subsamples of 3K items, mean±std). LLM view pairs lie in 0.82–0.94 **except Toys Function–Style (**.818**); TF-IDF vs. any LLM view stays within 0.18–0.20.**

View Pair	Beauty	Toys
Description–Function	.846 ± .001	.834 ± .001
Description–Audience	.860 ± .001	.856 ± .002
Description–Style	.914 ± .000	.904 ± .001
Function–Audience	.940 ± .000	.921 ± .001
Function–Style	.843 ± .001	.818 ± .002
Audience–Style	.859 ± .001	.838 ± .002
TF-IDF vs. LLM view (range)	.196–.201	.184–.189

Table 8: Effective rank (Beauty/Toys, 3 subsamples). 4-view concat occupies only 53–55% of its nominal dimensionality.

Representation	Dim	ER (Beauty)	ER (Toys)
Single LLM view (avg)	64	63.7 (99.5%)	63.6 (99.4%)
4-view concat	256	136.1 (**53.2%**)	141.2 (**55.2%**)
TF-IDF (SVD)	256	249.9 (97.6%)	251.4 (98.2%)

Table 9: Leave-one-view-out (LOVO; zero-mask, 3-seed mean) under GTS-Last warm, matching Table 1. Δ: warm MRR@10 change vs. the full 4-view model. Maximum |Δ| is 0.00032 (Style on Beauty).

Dataset	Full MRR	−Desc	−Func	−Aud	−Style
Beauty	.00636	−.00021	−.00030	−.00027	−.00032
Toys	.00417	−.00017	+.00005	−.00007	−.00025

(2) Pairwise cosine vs. CKA.. Mean pairwise cosine between prompt views is often moderate or near zero (e.g., Description–Function ≈ −0.01 on Beauty), while Function–Audience reaches ≈ 0.42. This does *not* contradict high CKA: cosine measures instance-level alignment of vectors, whereas CKA compares Gram-matrix geometry across the item set [11]. Views can have low average cosine yet still induce nearly isomorphic similarity structures, which is the regime we observe for prompt-based paraphrasing. We therefore treat CKA as the primary similarity diagnostic and use cosine only as a complementary check, never as standalone proof of redundancy.

(3) Low effective rank of concatenated views. Table 8 and Figure 2(b) show that concatenating four 64-d views into 256 dimensions yields effective rank of only 136 (Beauty) and 141 (Toys), occupying 53–55% of the available capacity. Nearly half the nominal dimensions carry redundant structure. TF-IDF at the same 256-d has effective rank ≥250 (97–98%), utilizing almost all capacity. Concatenating prompt views therefore expands the subspace far less than the dimension count suggests.

(4) Negligible LOVO impact. Table 9 shows that zeroing any single view has minimal downstream effect under GTS-Last warm: maximum |Δ| = 0.00032 (≈5% of the full TF+MV score). Full MRR

Table 10: Pre- vs. post-whitening audit (3 item-subsamples, mean). CKA is already ≥0.94 before whitening; whitening lowers CKA and raises concat effective rank, so redundancy is not a whitening artifact. Subsamples are drawn independently of Table 8, hence the ±1 difference in the Toys post-whitening rank.

Dataset	Stage	Pairwise CKA	Concat ER/256
Beauty	Pre-whiten (SVD+center)	0.94–0.99	99 (39%)
	Post-whiten (TS fit)	0.84–0.94	136 (53%)
Toys	Pre-whiten (SVD+center)	0.94–0.98	104 (41%)
	Post-whiten (TS fit)	0.82–0.92	140 (55%)

matches Table 1's TF+MV column. Per-seed signs can flip, so we do not treat any prompt as uniquely necessary.

Where is redundancy already present? To disentangle prompt-induced from preprocessing-induced redundancy, we compare CKA at two stages: *pre-whitening* (SVD-projected + centered, 64-d per view, without ZCA whitening) and *post-whitening* (the final model-input embeddings). On Beauty, mean pairwise CKA is 0.96 ± 0.001 pre-whitening and 0.88 ± 0.001 post-whitening; on Toys, 0.96 ± 0.001 and 0.87 ± 0.001 respectively. CKA is *already extremely high before whitening* (Table 10): whitening *reduces* mean inter-view CKA by 0.08 (Beauty) and 0.10 (Toys), and raises concatenated effective rank from 99/256 (39%) to 136/256 (53%) on Beauty and from 104/256 to 140/256 on Toys.

To trace redundancy further upstream, we compute inter-view CKA directly on the raw 3584-d Qwen hidden states *before any SVD projection.* Mean pairwise CKA at 3584-d is 0.961 on Beauty and 0.963 on Toys (3-trial random subsample of 3000 items, $\sigma < 0.002$). Redundancy is therefore already present in the raw Qwen outputs before SVD or whitening under our prompt design, rather than an artifact of SVD truncation. We cannot separate encoder geometry from the shared item text or from limited prompt divergence: even at this stage the four views yield near-isomorphic relational structures. Whitening partially decorrelates the views but cannot remove this structural overlap. A wider per-view projection is thus unlikely to remove the overlap (Table 10).

Synthesis. The three diagnostics agree: prompt views encode overlapping semantics, which produces dimensionality collapse and negligible per-view contribution. The observed geometry is consistent with insufficient view complementarity for views derived from a single source via semantic paraphrasing [24]. The potential complementarity in our system lies between lexical TF-IDF and semantic LLM views (CKA 0.18–0.20), suggesting representational dissimilarity that is not harvested by naive multi-prompt stacking: the TF+MV model replicates similar semantic structures four times instead of further exploiting the lexical–semantic axis it already contains.

7 Discussion and Limitations

Why does TF-IDF alone match TF+MV?. Under GTS-Random, TF-IDF matches or exceeds the four-view variant on warm targets (Table 1). First, TF-IDF captures *lexical diversity*: the vocabulary of product titles and descriptions spans tens of thousands of unique tokens, and SVD-projected TF-IDF retains nearly all available capacity (ER/dim ≈0.98; Table 8). Second, the four LLM views share highly similar relational geometry (CKA 0.82–0.94) because the same encoder processes the same item text with only minor instruction changes. Concatenating them nominally quadruples dimensionality but only about doubles the effective rank (64 → 136), so the concatenated views occupy about half of their nominal rank (Table 8). TF-IDF does not capture richer semantics than an LLM; its geometric diversity per dimension is simply higher, which is a practical advantage under GTS-Random.

LLM-only (no TF-IDF base) ablation. Because TF+LLM and TF+MV augment the TF-IDF base, one may ask how much the LLM views achieve on their own. We train a variant using only a single Qwen2.5-7B view with *no* TF-IDF base (same architecture, same two-phase pipeline). Under the main protocol (GTS-Last, warm targets, users with ≥ 5 training interactions, train-catalog candidates), this LLM-only variant scores .0046 ± .0001 on Beauty and .0018 ± .0007 on Toys—below the TF+LLM column in Table 1 (.0060 / .0035) and below ID-only on Toys. The lexical base is therefore load-bearing, not a "free init": all of our text gains are gains of TF-IDF-anchored pipelines.

Single-view hybrid: dropping three of the four views. To test whether the lexical–semantic dissimilarity identified in RQ3 (TF–LLM CKA 0.18–0.20) is what carries the useful signal, we train a hybrid that keeps the TF-IDF base and a *single* LLM view inside the multi-view architecture (per-view projection + concatenation; no extra prompt views), under the same cutoff-fit protocol and training seeds as Table 1. Under GTS-Last warm, the hybrid scores .0070 ± .0006 (Beauty) and .0049 ± .0011 (Toys), vs. TF+MV .0064 ± .0003 / .0042 ± .0006; under GTS-Random (3 target-sampling seeds) it scores .0128 ± .0003 and .0114 ± .0008, vs. TF+MV .01244 ± .0002 / .0108 ± .0009. The hybrid thus matches or slightly exceeds the four-view model on both datasets under both target modes while using one prompt view instead of four, with margins within one seed-level std—*no loss from dropping three views*, not a large additive gain. Low TF–LLM CKA by itself licenses only *geometric dissimilarity*; the hybrid provides the ranking-level evidence.

Implications.

(1) Report text-representation comparisons under GTS with explicit target mode.
(2) Start with cutoff-fit TF-IDF as a strong baseline. Under GTS-Random it stays within one std of TF+MV on all three datasets and leads on Toys; under GTS-Last on the Main window it may trail TF+LLM/TF+MV, but Table 4 shows that an earlier cutoff can reverse that ranking on Beauty. Given its negligible preprocessing cost relative to LLM inference, TF-IDF should be the first text baseline evaluated.
(3) A single LLM view typically matches four prompt views; diagnose redundancy (e.g., via CKA and effective rank) before adding more prompt views.
(4) Adding more prompt views from the same encoder yields negligible gains. Geometric diversity between representation families

Table 11: Recommendation-stage parameter count and online scoring cost (seed 42; *offline text encoding excluded*). TF+MV adds <1% parameters.

Model	Params (M)	ms/sample
ID / TF / TF+LLM / TF+MV	67.2 / 67.4 / 67.6 / 67.7	0.02
UniSRec	67.7	0.04

Beauty; Toys is ∼87M due to larger vocab (ranking identical).

(lexical TF-IDF vs. semantic LLM) is the more useful axis: in our runs, a TF-IDF + single-LLM-view hybrid matched or slightly exceeded TF+MV under both GTS-Last and GTS-Random.

Limitations.

- **Warm-target scope.** Our models lack content-only zero-shot fallback; warm-target evaluation is thus the fairest comparison, but it excludes 30–38% of eligible users' last test targets (and 30–41% of all post-cutoff targets).
- **Static metadata.** We use fixed product titles/descriptions; temporally evolving text (e.g., reviews, trending tags) could change the TF/LLM balance.
- **Amazon-only domain.** All three datasets are Amazon subcategories. Grocery is a same-platform replication, not a solution to domain diversity.
- **GTS-Random reproducibility.** Random is non-deterministic [4]; the 0.001–0.003 cross-seed std in Section 3.1 is measured on our checkpoints.
- **Redundancy locus.** Our redundancy audit spans three stages—raw 3584-d Qwen hidden states, 64-d SVD-projected pre-whitening, and 64-d post-whitening—showing that redundancy is already present in the raw Qwen outputs before SVD/whitening under our prompt design (Section 6). However, only a single LLM (Qwen2.5-7B) is tested, and we cannot isolate encoder geometry from shared item text or prompt similarity; other architectures or more divergent prompts may exhibit different inter-view structure.
- **Successive protocol.** Gusak et al. [4] identify Successive—scoring every test interaction in chronological order—as the closest offline analogue of online next-item prediction, and treat Last/Random as cheaper proxies with high rank-correlation to it. We report Last and Random; full Successive scoring at sampled-user scale remains future work.

Cost analysis. Table 11 shows that limited multi-view gains are not a recommendation-stage capacity issue: all variants have effectively the same parameter budget and online scoring cost. The practical cost gap lies in *offline text encoding*, which the table deliberately excludes: cutoff-fit TF-IDF +SVD takes about a minute on CPU for the Beauty catalog (259K items, measured), whereas each Qwen2.5-7B view requires a one-time GPU inference pass over the catalog, and multi-view multiplies that pass by the number of prompts. Feature-side cost, not model-side cost, is where TF-IDF, TF+LLM, and TF+MV differ.

8 Conclusion

We conducted a leakage-aware controlled study of text representations for sequential recommendation, structured as an empirical investigation rather than a model-proposal paper.

(1) **Protocol matters.** The relative ordering of TF-IDF, TF+LLM, and TF+MV depends on the GTS target mode. Text features consistently outperform ID-only, but which text is best requires specifying the evaluation protocol. Among the SASRec-based variants under GTS-Last, point estimates favor TF+MV on Beauty and Grocery, while Toys is TF+MV ≈ TF (the adapted UniSRec baseline is strongest on Grocery); under GTS-Random, TF-IDF catches up and overtakes it on Toys, with no gap exceeding one std. This shift is driven entirely by target selection, not by model changes.
(2) **No text variant is consistently dominant.** Under a matched-protocol equal-horizon Early window on Beauty (60% cutoff; the same 10% valid/test spans as Main, final 20% unused), absolute MRR drops and the Main ranking is not preserved (TF-IDF leads narrowly), yet text variants remain within 0.001 of each other.
(3) **Multi-view gains are limited by inter-view redundancy.** CKA, effective rank, and LOVO jointly show that prompt views encode overlapping semantics; potential complementarity lies between lexical and semantic features (CKA 0.18–0.20), not between different prompt instructions applied to the same encoder. Consistently, a hybrid that keeps the TF-IDF base and a single LLM view matches or slightly exceeds TF+MV under both GTS-Last and GTS-Random (Section 7).

The practical default under leakage-aware evaluation is: report GTS-Last (and Random as a check), keep cutoff-fit TF-IDF as a mandatory baseline, and treat multi-view LLM features as a hypothesis to be tested—not a free upgrade.

Future directions. Five questions remain open. (1) **Richer lexical–semantic fusion.** Whether more sophisticated fusion (e.g., gating, cross-attention) can turn the lexical–semantic pair into larger, significant gains remains open. (2) **Cross-architecture redundancy audit.** Our raw 3584-d audit finds this redundancy already in Qwen2.5-7B outputs under our prompt design. Testing whether the same inter-view collapse holds for decoder-only vs. encoder-only LLMs (e.g., Mistral, BERT-large) would clarify whether this is a general transformer property or specific to the Qwen family. (3) **GTS-Successive evaluation** at sampled-user scale, to validate Last/Random as proxies for online next-item prediction [4]. (4) **Non-Amazon domains** with reliable timestamps, to test whether protocol sensitivity generalizes beyond product catalogs. (5) **Instruction-diverse prompts** that enforce semantic divergence (e.g., attribute extraction vs. sentiment summarization) to test whether complementarity can be engineered at the prompt level.

Ethical Considerations

This work uses public Amazon review datasets with no personally identifiable attributes beyond standard user/item IDs. Potential concerns include reinforcing popularity bias under temporal shift

and inheriting biases from LLM-generated text features. We report warm-target stratified diagnostics. No human subjects were involved.

A GTS-All Results

Main tables use GTS-Last (one last target per user). GTS-All evaluates all post-cutoff targets (Beauty: 4,520 users → 12,001 samples, avg 2.65/user; Toys: 4,873 users → 13,186, avg 2.71/user).

Table 12: GTS-All warm MRR@10 (3-seed mean; cutoff-fit checkpoints for all columns). Beauty TF/TF+MV to 5 decimals because both round to .0241**. Bold: row max (same rule as Table 1).**

Dataset	ID	TF	TF+LLM	TF+MV	UniSRec
Beauty	.0156	.02411	.0237	**.02413**	.0180
Toys	.0118	**.0191**	.0180	.0184	.0139

Under GTS-All, TF leads TF+MV on Toys and trails it by .00002 on Beauty, closer to the GTS-Random picture than to GTS-Last, where TF+MV leads narrowly among the SASRec-based variants. All has the lowest rank-correlation with Successive and introduces task mismatch [4].

B Grocery Per-Seed Breakdown

Grocery replicates the Beauty/Toys pattern under the same search budget: among the SASRec-based variants, TF+MV leads narrowly on Last, and TF and TF+MV stay within one std of each other on Random.

Table 13: Grocery warm MRR@10 per training seed under GTS-Last and GTS-Random (Random averaged over target-sampling seeds $\{42, 123, 456\}$**). Bold: row max among ID/TF/TF+LLM/TF+MV.**

Protocol	Seed	ID	TF	TF+LLM	TF+MV
GTS-Last	42	.0027	.0038	.0049	**.0052**
	2024	.0019	.0049	.0050	**.0058**
	2026	.0043	.0050	.0041	**.0065**
GTS-Rand	42	.0044	.0075	.0073	**.0088**
	2024	.0041	**.0095**	.0080	.0082
	2026	.0063	.0091	.0075	**.0097**

C GTS-Random Target-Sampling Seeds

Main-table GTS-Random numbers average three target-sampling seeds $\{42, 123, 456\}$ on the same checkpoints (no retraining). Table 14 reports each seed after averaging the three training seeds. Cross-seed std of warm MRR@10 is typically 0.001–0.003; seed 456 is systematically lower, which is why a single Random seed would overstate TF's Beauty lead (0.0138 vs. 3-seed 0.01239).

Table 14: GTS-Random warm MRR@10 by target-sampling seed (mean over training seeds 42/2024/2026). Bold: row max.

Dataset	Sample seed	ID	TF	TF+LLM	TF+MV	UniSRec
Beauty	42	.0081	**.0138**	.0133	.0134	.0098
	123	.0080	.0134	.0128	**.0141**	.0103
	456	.0063	**.0101**	.0099	.0098	.0087
Toys	42	.0096	**.0146**	.0140	.0141	.0103
	123	.0066	**.0091**	.0089	.0085	.0078
	456	.0065	**.0101**	.0095	.0098	.0075

D Last vs. Random on Intersecting Users

Last and Random score the same eligible users, but the warm filter applies to the *selected* target, so the two warm sub-cohorts do not coincide exactly (Section 3.1). We re-score both modes on the *intersection* of their warm cohorts—users whose Last target *and* whose Random target both lie in the train catalog—using the same checkpoints and a single forward pass per checkpoint, so that per-user ranks are identical to those behind the main tables. The intersection retains 2,846–2,858 users on Beauty (63% of eligible), 2,651–2,686 on Toys (54–55%), and 2,671–2,678 on Grocery (64%), varying only with the Random sampling seed.

Table 15: Warm MRR@10 on intersecting users (warm under both Last and Random) vs. the full warm cohorts used in the main tables. Mean±std over 3 training seeds; Random additionally averages target-sampling seeds $\{42, 123, 456\}$**. Bold: highest among ID/TF-IDF/TF+LLM/TF+MV in that column (same rule as Table 1; UniSRec not in this table).**

		Last		Random	
Dataset	Variant	Intersect.	Full warm	Intersect.	Full warm
Beauty	ID	.0033±.0007	.0032±.0006	.0066±.0005	.0074±.0005
	TF-IDF	.0054±.0004	.0053±.0004	.0110±.0003	**.0124**±.0002
	TF+LLM	.0063±.0005	.0060±.0004	.0107±.0003	.0120±.0002
	TF+MV	**.0065**±.0004	**.0064**±.0003	**.0113**±.0002	**.0124**±.0002
Toys	ID	.0032±.0002	.0031±.0002	.0073±.0001	.0076±.0002
	TF-IDF	.0039±.0006	.0040±.0005	**.0110**±.0004	**.0113**±.0004
	TF+LLM	.0035±.0002	.0035±.0002	.0106±.0007	.0108±.0007
	TF+MV	**.0040**±.0005	**.0042**±.0006	.0105±.0009	.0108±.0009
Grocery	ID	.0031±.0012	.0030±.0012	.0047±.0012	.0050±.0012
	TF-IDF	.0049±.0007	.0046±.0006	.0083±.0012	.0087±.0011
	TF+LLM	.0050±.0005	.0047±.0005	.0071±.0002	.0076±.0004
	TF+MV	**.0063**±.0006	**.0058**±.0007	**.0087**±.0009	**.0089**±.0008

Table 15 reports warm MRR@10 on the intersection alongside the full warm cohorts of Tables 1 and 2. Two observations hold in all six dataset×mode blocks. First, restricting to the intersection changes absolute scores by at most .0014 (largest on Beauty Random, where dropping the Last-only warm users lowers TF from .0124 to .0110) and does not change which of TF-IDF and TF+MV leads. Second, the Last→Random shifts we rely on survive on the identical user set: point estimates favor TF+MV under Last on Beauty and Grocery, with Toys TF+MV ≈ TF, whereas under Random TF-IDF closes to within one std on Beauty (.0110 vs. .0113) and Grocery (.0083 vs. .0087) and overtakes TF+MV on Toys (.0110 vs. .0105). Random remains easier than Last for every variant.

E UniSRec Detailed Results

UniSRec [8] is protocol-sensitive even under the shared filter (users with at least 5 training interactions; same cutoff-fit checkpoints):

Table 16: UniSRec warm MRR@10 under the shared filter (3-seed mean±std).

Dataset	GTS-All	GTS-Last	GTS-Random†
Beauty	.0180±.0001	.0052±.0002	.0096±.0001
Toys	.0139±.0003	.0040±.0008	.0085±.0003

†GTS-Random averaged over target-sampling seeds {42, 123, 456} (same protocol as Table 1). All three columns are warm-target MRR@10 on the same cutoff-fit checkpoints.

References

[1] Keqin Bao, Jizhi Zhang, Yang Zhang, Wenjie Wang, Fuli Feng, and Xiangnan He. 2023. TALLRec: An Effective and Efficient Tuning Framework to Align Large Language Model with Recommendation. In *Proceedings of the 17th ACM Conference on Recommender Systems (RecSys)*. Singapore, 1007–1014. doi:10.1145/3604915.3608857

[2] Kawin Ethayarajh. 2019. How Contextual are Contextualized Word Representations? Comparing the Geometry of BERT, ELMo, and GPT-2 Embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. 55–65. doi:10.18653/v1/D19-1006

[3] Shijie Geng, Shuchang Liu, Zuohui Fu, Yingqiang Ge, and Yongfeng Zhang. 2022. Recommendation as Language Processing (RLP): A Unified Pretrain, Personalized Prompt & Predict Paradigm (P5). In *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys)*. Seattle, WA, USA, 299–315. doi:10.1145/3523227.3546767

[4] Danil Gusak, Anna Volodkevich, Anton Klenitskiy, Alexey Vasilev, and Evgeny Frolov. 2025. Time to Split: Exploring Data Splitting Strategies for Offline Evaluation of Sequential Recommenders. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems*. 874–883. doi:10.1145/3705328.3748164

[5] Balázs Hidasi and Alexandros Karatzoglou. 2018. Recurrent Neural Networks with Top-k Gains for Session-based Recommendations. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management (CIKM)*. 843–852. doi:10.1145/3269206.3271761

[6] Balázs Hidasi, Alexandros Karatzoglou, Linas Baltrunas, and Domonkos Tikk. 2016. Session-based Recommendations with Recurrent Neural Networks. In *Proceedings of the 4th International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/1511.06939

[7] Yupeng Hou, Zhankui He, Julian McAuley, and Wayne Xin Zhao. 2023. Learning Vector-Quantized Item Representation for Transferable Sequential Recommenders. In *Proceedings of the ACM Web Conference 2023 (WWW)*. Austin, TX, USA, 1162–1171. doi:10.1145/3543507.3583434

[8] Yupeng Hou, Shanlei Mu, Wayne Xin Zhao, Yaliang Li, Bolin Ding, and Ji-Rong Wen. 2022. Towards Universal Sequence Representation Learning for Recommender Systems. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. Washington, DC, USA, 585–593. doi:10.1145/3534678.3539381

[9] Yitong Ji, Aixin Sun, Jie Zhang, and Chenliang Li. 2023. A Critical Study on Data Leakage in Recommender System Offline Evaluation. *ACM Transactions on Information Systems* 41, 3 (2023), 75:1–75:27. doi:10.1145/3569930

[10] Wang-Cheng Kang and Julian McAuley. 2018. Self-Attentive Sequential Recommendation. In *Proceedings of the 2018 IEEE International Conference on Data Mining (ICDM)*. 197–206. doi:10.1109/ICDM.2018.00035

[11] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of Neural Network Representations Revisited. In *Proceedings of the 36th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 97)*. 3519–3529.

[12] Walid Krichene and Steffen Rendle. 2020. On Sampled Metrics for Item Recommendation. In *Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. 1748–1757. doi:10.1145/3394486.3403226

[13] Jiacheng Li, Ming Wang, Jin Li, Jinmiao Fu, Xin Shen, Jingbo Shang, and Julian McAuley. 2023. Text Is All You Need: Learning Language Representations for Sequential Recommendation. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. 1258–1267. doi:10.1145/3580305.3599519

[14] Julian McAuley, Christopher Targett, Qinfeng Shi, and Anton van den Hengel. 2015. Image-based Recommendations on Styles and Substitutes. In *Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. 43–52. doi:10.1145/2766462.2767755

[15] Zaiqiao Meng, Richard McCreadie, Craig Macdonald, and Iadh Ounis. 2020. Exploring Data Splitting Strategies for the Evaluation of Recommendation Models. In *Proceedings of the 14th ACM Conference on Recommender Systems (RecSys)*. 681–686. doi:10.1145/3383313.3418479

[16] Jianmo Ni, Jiacheng Li, and Julian McAuley. 2019. Justifying Recommendations using Distantly-Labeled Reviews and Fine-Grained Aspects. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 188–197. doi:10.18653/v1/D19-1018

[17] Qwen Team. 2024. Qwen2.5 Technical Report. *arXiv preprint arXiv:2412.15115* (2024).

[18] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. 2009. BPR: Bayesian Personalized Ranking from Implicit Feedback. In *Proceedings of the 25th Conference on Uncertainty in Artificial Intelligence (UAI)*. 452–461.

[19] Olivier Roy and Martin Vetterli. 2007. The effective rank: A measure of effective dimensionality. In *2007 15th European signal processing conference*. IEEE, 606–610.

[20] Aixin Sun. 2023. Take a Fresh Look at Recommender Systems from an Evaluation Standpoint. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. 2629–2638. doi:10.1145/3539618.3591931

[21] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. 2019. BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*. 1441–1450. doi:10.1145/3357384.3357895

[22] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2018. Representation Learning with Contrastive Predictive Coding. *arXiv preprint arXiv:1807.03748* (2018). doi:10.48550/arXiv.1807.03748

[23] Hao Wang, Naiyan Wang, and Dit-Yan Yeung. 2015. Collaborative Deep Learning for Recommender Systems. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 1235–1244. doi:10.1145/2783258.2783273

[24] Chang Xu, Dacheng Tao, and Chao Xu. 2013. A Survey on Multi-view Learning. *arXiv preprint arXiv:1304.5634* (2013). doi:10.48550/arXiv.1304.5634

[25] Tingting Zhang, Pengpeng Zhao, Yanchi Liu, Victor S. Sheng, Jiajie Xu, Deqing Wang, Guanfeng Liu, and Xiaofang Zhou. 2019. Feature-level Deeper Self-Attention Network for Sequential Recommendation. In *Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI)*. 4320–4326. doi:10.24963/ijcai.2019/600

[26] Wayne Xin Zhao, Shanlei Mu, Yupeng Hou, Zihan Lin, Yushuo Chen, Xingyu Pan, Kaiyuan Li, Yujie Lu, Hui Wang, Changxin Tian, Yingqian Min, Zhichao Feng, Xinyan Fan, Xu Chen, Pengfei Wang, Wendi Ji, Yaliang Li, Xiaoling Wang, and Ji-Rong Wen. 2021. RecBole: Towards a Unified, Comprehensive and Efficient Framework for Recommendation Algorithms. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*. 4653–4664. doi:10.1145/3459637.3482016